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DSA

- Q11) Ans → b) Data & a pointer to the next node.
- Q12) Ans → c) Null
- Q13) Ans → b) Self-referential structure
- Q14) Ans → b) head == Null
- Q15) Ans → b) newnode → next = head; head = new node;
- Q16) Ans → b) temp = head; head = head → Next; free (temp);
- Q17) Ans → c) Traversing from tail to head
- Q18) Ans → b) n
- Q19) Ans → c) copying the data of P → next into P, then deleting P → next
- Q20) Ans → b) while (P != Null)
- Q21) Ans → b) P → Next == NULL
- Q22) Ans → b) A stack of visited nodes
- Q23) Ans → c) Three
- Q24) Ans → b) Exactly one node
- Q25) Ans → b) P = (struct node *) malloc (size of (struct node));
- Q26) Ans → b) Avoid special-case handling for insertion & deletion at the head
- Q27) Ans → c) The first node
- Q28) Ans → b) It does not support random (indexed) access
- Q29) Ans → b) Loss of access to all nodes, causing a memory leak
- Q30) Ans → b) A slow pointer & a fast pointer
- Q31) Ans → c) The last node
- Q32) Ans → b) Nodes may be scattered anywhere in memory
- Q33) Ans → struct node {
 int data;
 struct node * next;
};

Q24) Ans:- The code prints the sum of each node & its next node;

$$\rightarrow 10+20=30$$

$$\rightarrow 20+30=50$$

$$\rightarrow 30+40=70$$

Output is $\rightarrow 305070$.

Q25) Ans:- Error:- head is passing by value, so changes to the head pointer not affect the original list. Also, an empty list case needs handling correct version:-

```
void insert End (struct node ** head; int n)
{
    struct node * t = (struct node *) malloc (sizeof(struct node));
```

```
    t->data = n;
```

```
    t->next = NULL;
```

```
    if (*head == NULL) {
```

```
        *head = t;
```

```
        return;
```

```
    }
```

```
    struct node * p = *head;
```

```
    while (p->next != NULL) {
```

```
        p = p->next;
```

```
        p->next = t;
```

```
    }
```

Q26) Ans:- struct node * reverse (struct node * head)

```
{
    struct node * prev = NULL;
```

```
    struct node * curr = head;
```

```
    struct node * Next = NULL;
```

```
    while (curr != NULL) {
```

```
        Next = curr->next;
```

```
        curr->next = prev;
```

```
        prev = curr;
```

```
        curr = Next;
```

```
    }
```

```
    return prev;
```

```
}
```

Q27 Ans Suppose the list is $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

$\Rightarrow$ I want to delete 30.

$\rightarrow$ then, first 20 nodes next pointer needs to be pointer to 40.

Before $\rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$
prev delete next

After $\rightarrow 10 \rightarrow 20 \rightarrow \uparrow \rightarrow 40 \rightarrow \text{NULL}$
30 deleted

prev $\rightarrow$ Next = del $\rightarrow$ next ;
free (del) ;

Q28

Ans Initial list :-

$5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$

i) $7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{NULL}$

ii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow \text{NULL}$

iii) $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$

final answer :- $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{NULL}$